EXNO:5
19-2-26

Classification using K-Nearest (KNN) and decision Tree

Pre lab

1) what is the intuition behind the K-Nearest Neighbors (KNN) algorithm?

KNN works on the idea that similar data points exist close to each other. A new point is classified based on the majority class of its K nearest neighbors in the feature space.

2) why is KNN called a "lazy learner"?

KNN is called a lazy learner because it does not build a model during trainning. It simply stores the training data and performs computation only at prediction time

3) How does the choice of K influence model performance?

* small K → low bias, high variance (may overfit)
* large K → high bias, low variance (may underfit)

choosing K balances the bias-variance tradeoff.

4) why is feature scaling important in KNN?

KNN uses distance measures (like Euclidean distance) of features are on different scales, large-scale features dominate the distance calculation. Feature scaling ensures fair contribution from all features

5) what is decision Tree classifies?

A decision Tree is a supervised learning algorithm that splits data into subsets based on feature conditions forming a tree-like structure to make classification decisions

6) what is meant by tree depth?

Tree depth is the maximum number of levels from the root node to the deepest leaf node. greater depth means more complex decision boundaries.

7) what is overfitting in decision tree?

overfitting occurs when the tree becomes too deep and memorizes training data, including noise, leading to poor generalization on new data.

8 Difference b/w Entropy and gini index?

Entropy	gini index
Based on information theory	Based on impurity measure
uses logarithmic calculation	Simpler mathmatical formula
Range 0 to 1	Range 0 to 0.5
used in ID3	used in CART

Post lab:-

Scenario 0-1 K-Nearest Neighbors (Breast cancer dataset)

1) why was feature scaling necessary?

KNN uses distance calculations (Euclidean distance) Features like area and perimeter have larger numerical range than smoothness, so without scaling they would dominate the distance metric scaling ensured equal contribution of all medical features.

2) How did model performance change with k?

* small k → very high trainning accuracy, but unstable test accuracy
* Moderate k → Balanced and stable performance
* Large k → Acuracy decreased (underfitting)

3) what value of k gives best results? why?

Typically k = 5 or 7 gave best result because it balanced

* low noise sensitivity
* good generalization
* stable decision boundary

it reduced overfitting compared to k = 1

4) what are limitations of KNN?

* computationally expensive for large datasets
* sensitive to irrelevant features
* Requires feature scaling
* performance drops in high dimensions (curse of dimensionality)

Scenario-2 decision Tree
(Loan prediction dataset)

1) which features influenced loan approval most?

Typically:

* credit history (most important)
* Application income
* loan amount

Credit history strongly influences approval because it directly reflect repayment reliability

2) How did tree depth affect performance?

* shallow tree → simpler model, may underfit
* deep tree → higher training accuracy but risk of overfitting.

Optimal depth improved test accuracy while avoiding memorization.

3) what signs of overfitting were observed?

* Very high training accuracy
* lower testing accuracy
* highly complex tree structure
* May small leaf nodes.

4) Advantage of decision tree?

* easy to interpret
* No need for feature scaling
* handles categorical data directly
* provides feature importance

1 compare KNN and decision Tree?

KNN	decision Tree
lazy learner	Eager learner
Distance-based	Rule-based splitting
Needs scaling	No scaling needed
slow prediction	Fast prediction

2) which model is more interpretable? why?

Decision tree is more interpretable because it produces clear if-else rules and a visual tree structure

3) which model is more sensitive to noise?

* KNN is sensitive when k is small
* Decision Tree are sensitive when depth is large. however, KNN is generally more affected by noise data points

4) why is hyperparameter tuning important?

hyperparameter like

* k in KNN
* depth in decision Tree

Control model complexity and help acheive optimal bias variance balance.

5) why is confusion matrix preferred over accuracy alone?

Accuracy does not show type of errors confusion matrix provides

* True positives
* True Negatives
* False positive
* False Negative

This is crucial in medical diagnosis and loan approval scenarios

Description	Max mark	Obtained mark
Pre lab	5	3
In lab	10	10
Post lab	5	3
viva	10	3
Total	30	19
Faculty signature		